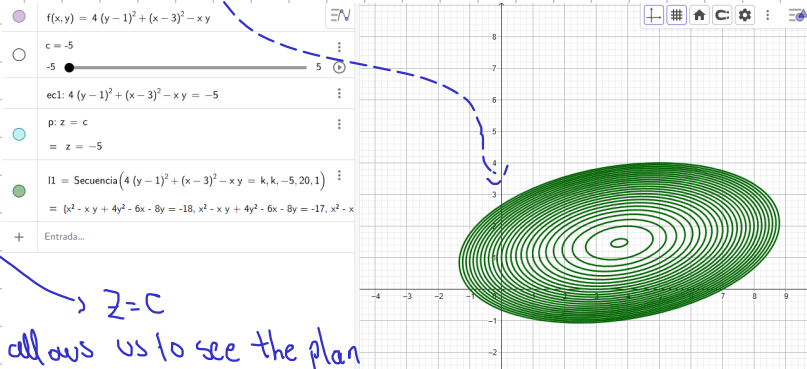
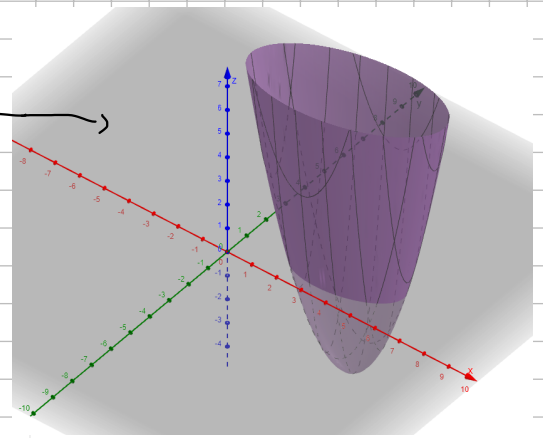


Exact Diff Equations

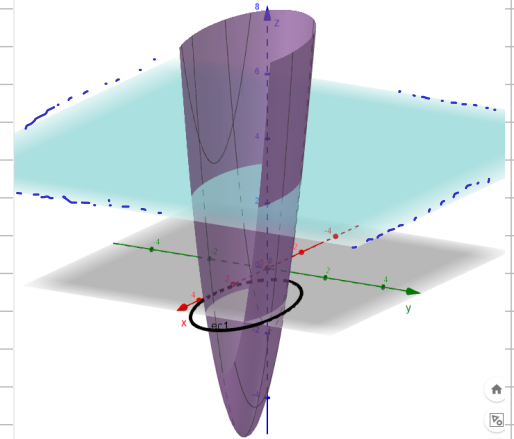
$$F: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$(x,y) \mapsto F(x,y) = 4(y-1)^2 + (x-3)^2 - xy$$

• level curve: $CL_F(K) = \{(x,y) \in \mathbb{R}^2 / F(x,y) = K\}$
(when the f is isolated to c)



$z=c$
allows us to see the plan



$$C_{lf}^3(t): [a,b] \subset \mathbb{R} \rightarrow \mathbb{R}^2$$

$$t \mapsto C_{lf}^3(t) = (x(t), y(t))$$

$$\nabla f(x,y) = \left(\frac{\partial F}{\partial x}(x,y), \frac{\partial F}{\partial y}(x,y) \right)$$

$$= (2(x-3)-y, 8(y-1)-x)$$

$$h(t) = F \circ C_{lf}^3$$

$$= F(C_{lf}^3(t))$$

$$= 3$$

$$\frac{dh}{dt}(t) = 0$$

$$\frac{dh}{dt} = \frac{d}{dt} [f(C_{lf}^3(t))] = \nabla F(C_{lf}^3(t)) \cdot \frac{dC_{lf}^3(t)}{dt}$$

$$C_{lf}^3(t): t \mapsto C_{lf}^3(t) = (x(t), y(t))$$

$$\frac{dC_{lf}^3(t)}{dt} = \left(\frac{dx}{dt}(t), \frac{dy}{dt}(t) \right)$$

